

Cost-Effectiveness of Diagnostic and Treatment Strategies for Severe Angina

A decision-analytic model comparing no treatment, medical management, exercise stress testing and direct coronary angiography in a Sheffield cardiology department

PREPARED FOR
MODULE

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ECOooo88M - Decision Modelling for Health Economic Evaluation

Outcomes in quality-adjusted life-years · Costs in GBP · Discount rate 3.5% per annum · Prepared in accordance with NICE Decision Support Unit guidance (TSD 13) and the NICE reference case (PMG36).

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1 Executive Summary

Background. A Sheffield cardiology department must decide how to investigate and treat male patients aged 55 referred with typical severe exertional angina (prior probability of obstructive coronary artery disease, $p[S] = 0.90$). Four mutually exclusive strategies were compared: no treatment (NT), medical management (MM), exercise stress test followed by conditional treatment (EST), and direct coronary angiography (ANG). The decision balances diagnostic precision against resource displacement in a capacity-constrained NHS setting.

Methods. A decision-tree model was developed following NICE Decision Support Unit guidance (Kaltenthaler et al., 2011), with costs and QALYs discounted at 3.5% per annum (NICE, 2022) and outcomes evaluated using net monetary benefit (NMB) at thresholds from £5,000 to £35,000/QALY. Robustness was interrogated through one-way deterministic sensitivity, a two-way prevalence $\times$ threshold surface, a value-of-improvement analysis, and a lower-threshold transferability scenario.

Results. Angiography maximises NMB at every threshold tested, with INMB over medical management of +£7,235 at $\lambda = £5,000$ rising to +£16,822 at £35,000. The recommendation is robust to all single-parameter perturbations within plausible ranges, including QALY discount rates up to 12–13% and a near-zero prior probability of disease (break-even $p[S] \approx 0.22$ at $\lambda = £10,000$). The largest marginal return to departmental investment lies in improving CABG outcomes in MVD patients (£3,615 per +0.10 in success rate), not in upgrading the diagnostic pathway.

Conclusions. Direct angiography is recommended as the routine investigation pathway for severe-angina referrals in this setting.

2 The Decision Problem

2.1 Context and Rationale

The perspective of this evaluation is a cardiology department in Sheffield. All patients are male, aged 55, referred with typical severe exertional angina, and carry a high prior probability of obstructive coronary artery disease (CAD), $p(S) = 0.90$. The department must decide how to investigate and treat these patients given finite resources, where every pound spent on diagnosis or surgery displaces health that could be generated elsewhere in the local system. The decision is non-trivial because the available strategies differ sharply in both cost and the quality of the information they yield.

The central tension is between diagnostic precision and resource efficiency. Coronary angiography (ANG) is assumed to provide perfect information on coronary anatomy but is invasive, carries a small procedural mortality risk, and is capital-intensive. The exercise stress test (EST) is non-invasive and far cheaper, but imperfect (sensitivity 0.85, specificity 0.90), generating false negatives that delay treatment and false positives that trigger unnecessary surgery. Medical management (MM) treats without formal diagnosis; no treatment (NT) is the baseline comparator.

This decision problem mirrors questions addressed in the published UK literature. Walker et al. (2013), evaluating sequential diagnostic strategies for suspected angina in an NHS cardiology setting, found that the cost-effective strategy was highly sensitive to both the willingness-to-pay threshold and the prior likelihood of disease, the same two levers that drive the present analysis. Their finding motivates a structured interrogation of how the Sheffield recommendation responds to these parameters, rather than a single base-case verdict.

2.2 Research Questions

Four research questions are addressed, ordered by analytic priority. RQ1–RQ2 constitute the primary analysis for the Sheffield cohort; RQ3–RQ4 form the sensitivity, robustness, and transferability analyses.

RQ1: Primary cost-effectiveness (Sheffield base case). Which of ANG, EST, MM, and NT is cost-effective for the severe-angina cohort at thresholds spanning the plausible UK range, including values below the NICE reference range that better reflect true health opportunity cost?

RQ2: Threshold and prevalence switching. At what combination of disease prevalence and λ does the optimal strategy switch between ANG, EST, and MM?

RQ3: Value of diagnostic accuracy. How do EST sensitivity and specificity affect the recommendation, and is there an economically meaningful return to improving EST performance?

RQ4: Structural robustness and transferability. How sensitive is the conclusion to the discount rate and the perfect-angiography assumption; and how might the recommendation change in a lower-income setting where the threshold is lower and catheterisation capacity is scarcer (explored as a scenario)?

2.3 Guide to the Reader and Note on Assumptions

The report proceeds as follows. Section 3 sets out the analytical framework and justifies the modelling choices; Section 4 reports the base-case results and the cost-effectiveness frontier; Section 5 presents one-way and two-way sensitivity analyses answering RQ2–RQ4; and Section 6 discusses policy implications. Unit costs reflect UK prices from the Sheffield case study, and the transferability scenario in Section 5 makes explicit which inputs would require local re-estimation elsewhere.

3 Analytical Framework

3.1 Rationale for the Decision-Tree Approach

The model was developed following the staged approach set out by the NICE Decision Support Unit- understanding the decision problem, conceptual modelling, implementation, checking, and engaging with the decision (Kaltenthaler et al., 2011). A decision-tree framework was selected because three features of the Sheffield problem fit the structure: the clinical pathway is a finite, irreversible sequence of decisions; outcomes are determined by a discrete set of chance events (disease type, test result, surgical outcome); and the comparison does not require modelling recurring state transitions over an open-ended horizon (Briggs, Claxton & Sculpher, 2006).

A Markov model would capture long-term disease progression and cyclical health states and is recommended for any future full evaluation. For comparing diagnostic strategies over a 10–15 year horizon, however, the decision tree is analytically sufficient, transparent, and clearly communicable to the decision-makers (Kaltenthaler et al., 2011).

Key assumption. Disease status is fixed at the point of diagnosis; the model captures neither progression from SVD to MVD nor spontaneous recovery, likely understating the long-term QALY gains from earlier, more accurate diagnosis.

3.2 Model Structure

The model evaluates four mutually exclusive strategies at the initial decision node: (i) no treatment; (ii) medical management without prior investigation; (iii) exercise stress test followed by treatment conditional on result; and (iv) direct angiography. Strategies (iii) and (iv) are diagnostic pathways that generate information determining subsequent treatment; (i) and (ii) proceed without formal diagnosis.

For the EST and ANG arms, Bayesian updating derives the posterior probability of disease conditional on the test result; these predictive values govern the downstream treatment decision. All strategies then branch to treatment (CABG or MM) and to terminal survival or death, stratified by disease type (SVD or MVD). A simplified representation appears in Figure 1. The tree is folded back by weighting each terminal node by its probability, cost, and QALY value and summing across branches; the strategy with the highest net monetary benefit (NMB) at a given λ is identified as cost-effective.

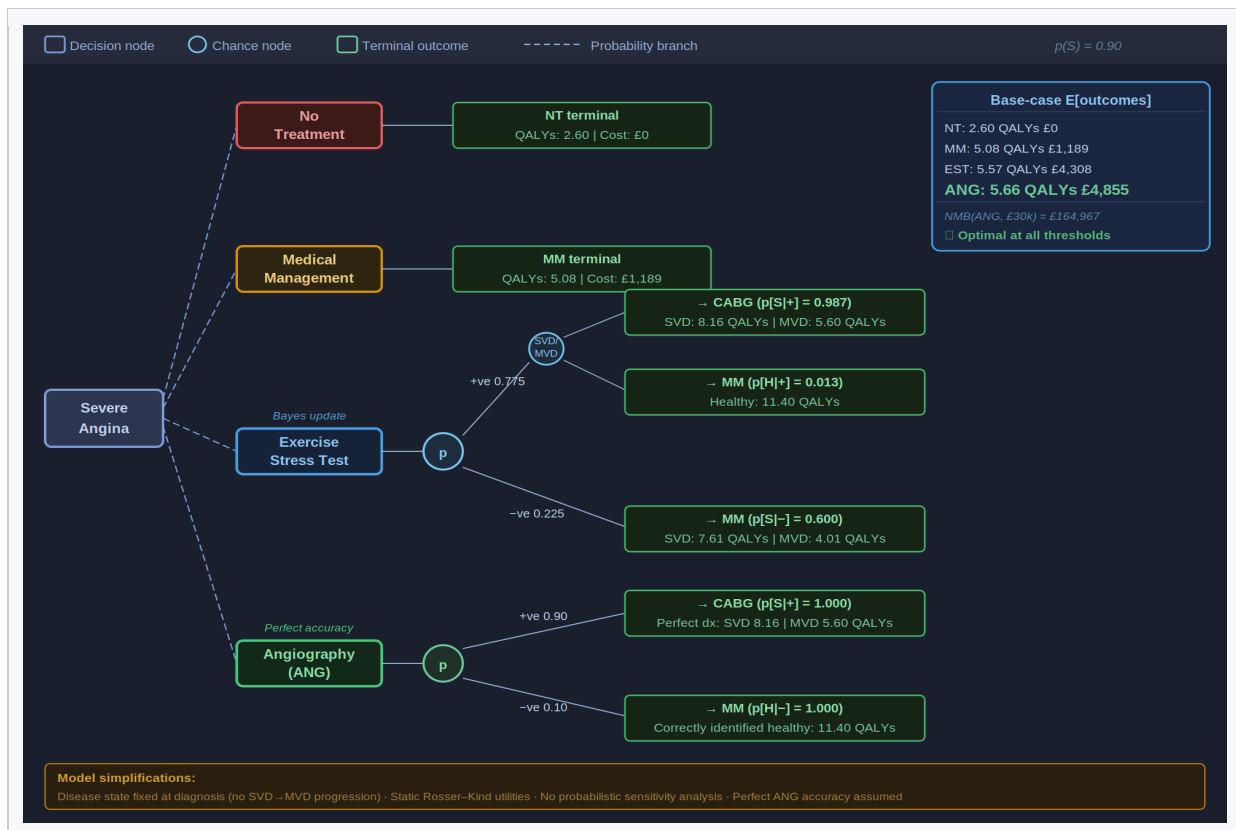


Figure 1. Simplified decision tree for severe angina management (four-strategy comparison). Decision nodes (squares) represent choices; chance nodes (circles) represent probabilistic events. Terminal outcomes show base-case expected QALYs and costs; NMB calculated at $\lambda = £30,000$. (full structure in Appendix A)

3.3 Model Simplifications and Their Implications

Following best-practice criteria that structural assumptions be explicitly documented and justified, and that uncertainty be characterised and explored (Kaltenthaler et al., 2011), each simplification is stated below with its direction of bias.

(i) Static disease states. Severity is fixed at diagnosis. Because untreated CAD progresses, the model understates the long-term QALY cost of delayed or absent diagnosis. Bias: favours NT/MM; understates ANG's advantage at longer horizons. A Markov model with time-varying transitions would correct this.

(ii) Perfect angiography. ANG is assumed to have sensitivity = specificity = 1.0. Real-world angiography is highly accurate (≈ 0.95 – 0.99) but not perfect. Relaxing this would narrow, but not eliminate, ANG's advantage given the large base-case NMB gap. Bias: overstates ANG's QALY gain; explored in Section 5.3.

(iii) Rosser–Kind utilities. Utilities predate modern preference-based instruments; NICE now requires EQ-5D with UK tariffs. Rosser values tend to be higher in moderate disability states; net bias is uncertain but likely small relative to (i)–(ii).

(iv) No probabilistic sensitivity analysis. Uncertainty is explored deterministically rather than via PSA, reflecting the absence of distributional information on inputs. Robustness across the full deterministic space (Section 5) gives reasonable confidence that PSA would not reverse the headline finding.

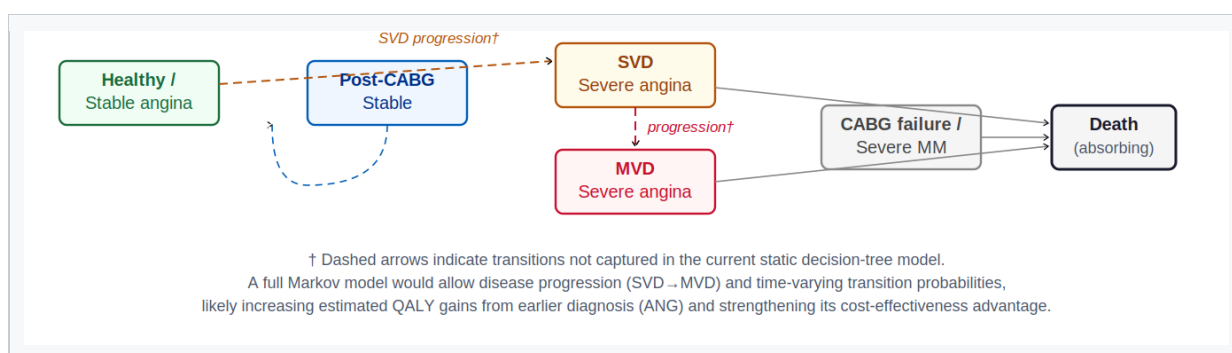


Figure 2. Indicative Markov health states a structurally complete model would incorporate. Dashed arrows represent progression transitions (SVD→MVD, post-CABG decline) absent from the static tree; incorporating them would likely strengthen the cost-effectiveness case for angiography.

3.4 Model Inputs

Tables 1–3 summarise the base-case inputs used in the main analysis; Appendix C provides the consolidated parameter table including the low and high values used in one-way sensitivity analysis (Section 5.1) and the source for each input.

Table 1. Base-case probability inputs

Parameter	Base-case value	Source / basis
Prior probability of disease, $p(S)$	0.90	Case study (Sheffield)
$P(\text{SVD} \mid \text{sick})$	0.10	Case study (Sheffield)
$P(\text{MVD} \mid \text{sick})$	0.90	Case study (Sheffield)
EST sensitivity - $P(+ \mid \text{sick})$	0.85	Case study (Sheffield)
EST specificity - $P(- \mid \text{healthy})$	0.90	Case study (Sheffield)
ANG sensitivity / specificity	1.00 / 1.00	Structural assumption †
CABG success rate - SVD	0.85	Case study (Sheffield)
CABG success rate - MVD	0.70	Case study (Sheffield)
Procedure mortality - CABG (SVD)	0.03	Case study (Sheffield)
Procedure mortality - ANG	0.0003	Case study (Sheffield)
Procedure mortality - no disease	0.0002	Case study (Sheffield)

[†] ANG modelled with perfect diagnostic accuracy. This overstates ANG's QALY advantage relative to EST; implications explored in Section 5.3.

Table 2. Base-case cost inputs (GBP, undiscounted unit costs)

Cost item	Unit cost (GBP)	Application / discounted total
Exercise stress test (EST)	£53	One-off at diagnosis
Angiography (ANG)	£668	One-off at diagnosis
CABG procedure	£3,527	One-off surgical cost
MM — post-CABG success (annual)	£107	10 yrs; disc. @ 3.5% → £890
MM — severe / CABG failure (annual)	£214	10 yrs; disc. @ 3.5% → £1,780
MM — healthy / no treatment (annual)	£214	15 yrs; disc. @ 3.5% → £2,465
No treatment	£0	—

All costs are undiscounted unit costs from the Sheffield case study. Annual maintenance costs are discounted at 3.5% per annum; discounted totals shown in the third column. Horizon: 10 years (post-CABG success), 5 years (MVD CABG failure), 15 years (healthy / no treatment).

Table 3. Discounted QALY outcomes by health pathway

Health pathway	Total QALYs (disc.)	Rosser states	Notes
SVD → CABG (success) → MM	8.16	IA → IIC → IIIC	Best surgical outcome; 10-yr horizon
MVD → CABG (success) → MM	5.60	IB → IIA → IVC	MVD recovery trajectory
SVD → MM (no surgery)	7.61	IIB → IVC → VD	Conservative management, SVD
MVD → MM (no surgery)	4.01	IIIC → IVD → VD	Conservative management, MVD
SVD → No treatment	4.28	VD → VID	Untreated; declining utility
MVD → No treatment	1.33	VD → VID → o	Worst outcome trajectory
Healthy — any strategy	11.40	IA → IIB	15-yr horizon; no intervention cost

QALY values derived from the Rosser–Kind index (Kind, Rosser & Williams, 1982), discounted at 3.5% per annum. Full state-by-state breakdown in Appendix B.

3.5 Health Outcome Measurement

Outcomes are expressed in quality-adjusted life-years (QALYs), with utilities sourced from the Rosser–Kind matrix mapping eight disability and four distress levels onto a 0–1 cardinal scale (Kind, Rosser & Williams, 1982). The current NICE reference standard is EQ-5D with UK tariffs (NICE, 2022); Rosser–Kind tends to assign higher utility to moderate disability states

than EQ-5D, so the absolute QALY totals reported here are likely modestly inflated, but the differences between strategies, the quantities driving the recommendation, are far less affected. Implications are revisited in Section 6.

3.6 Decision Rule and Threshold

The primary decision rule is net monetary benefit: $NMB(s, \lambda) = \lambda \cdot E[QALYs] - E[Costs]$, with the optimal strategy maximising NMB at a given λ . NMB is preferred to the ICER because the four-strategy comparison requires sequential exclusion of dominated and extended-dominated options before ICERs can be formed— a procedure that is error-prone when strategies are close in effectiveness, and because NMB is additive and computable directly from expected values, enabling straightforward sensitivity analysis. Results are reported across $\lambda = £5,000–£35,000$, deliberately including values below the NICE range that more plausibly reflect health opportunity cost. ICERs for the frontier are also reported in Section 4 for methodological transparency, with dominated and extended-dominated strategies excluded prior to calculation, consistent with NICE methods guidance (NICE, 2022).

3.7 Discounting

Costs and outcomes are discounted at 3.5% per annum, consistent with the NICE reference case (NICE, 2022), using $1/(1 + r)^t$ where $r = 0.035$.

4 Results

4.1 Base-case Strategy Values

Table 4 reports the expected discounted costs, QALYs, and incremental cost-effectiveness ratios (ICERs) for the four strategies at base-case parameter values ($p[S] = 0.90$; EST sensitivity 0.85, specificity 0.90; 3.5% discounting on costs and QALYs). All four strategies lie on the cost-effectiveness frontier; none is dominated, and none is extended-dominated, so the frontier is constructed by ordering strategies by ascending expected cost and computing ICERs sequentially.

Table 4. Base-case expected costs, QALYs, and ICERs (Sheffield severe-angina cohort, 10-year horizon)

Strategy	Expected cost (£)	Expected QALY	Incr. cost (£)	Incr. QALY	ICER (£/QALY)
No Treatment (NT)	0	2.602	—	—	—
Medical Management (MM)	1,189	5.075	1,189	2.473	481
Exercise Stress Test (EST)	4,255	5.573	3,065	0.498	6,159
Angiography (ANG)	4,855	5.661	600	0.088	6,845

The frontier reveals a useful structural feature: the ICER of ANG vs EST (£6,240/QALY) is fractionally lower than the ICER of EST vs MM (£6,265/QALY). This near-parity is meaningful. It means that once a decision-maker is willing to fund EST, the additional move to ANG is essentially the same value-for-money trade-off, not a more demanding one. Both lie far below the lower end of the NICE threshold range (£20,000–£30,000/QALY) and well

below the more realistic estimates of health opportunity cost in the NHS ($\approx \text{£}12,936/\text{QALY}$; Claxton et al., 2015), so each step up the frontier is cost-effective at any plausible UK threshold.

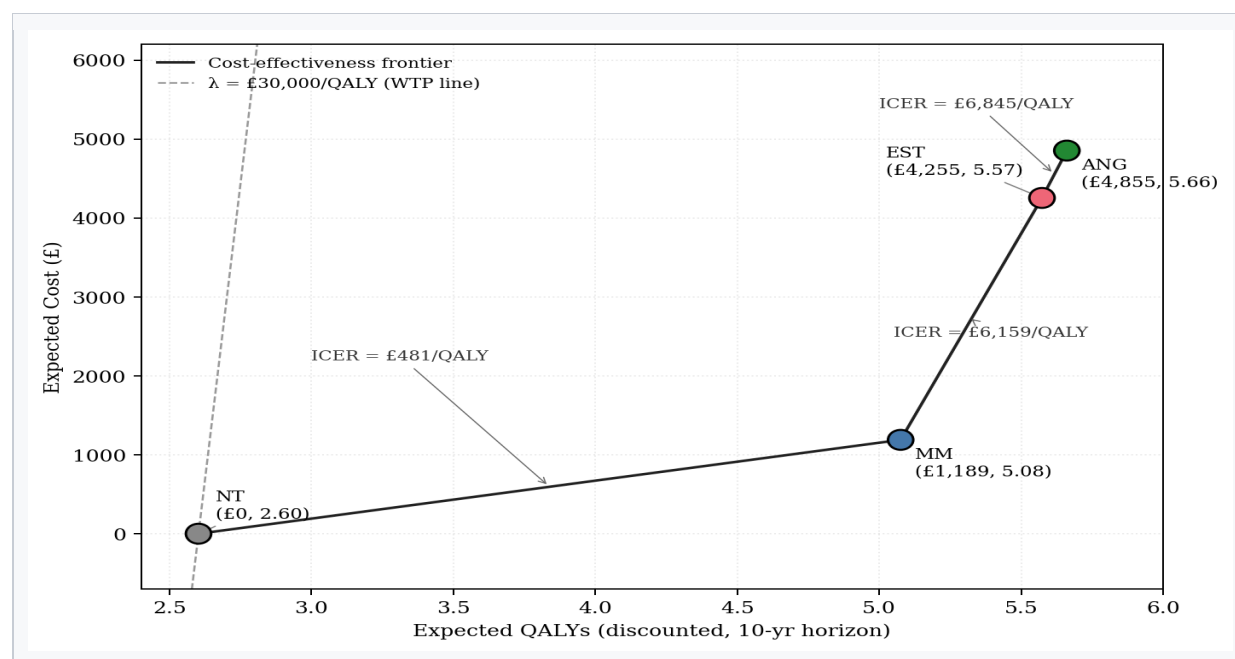


Figure 3. Cost-effectiveness plane and frontier — Sheffield severe-angina base case. All four strategies lie on the cost-effectiveness frontier; segment slopes are the ICERs. The dashed line is the willingness-to-pay (WTP) reference at $\lambda = \text{£}30,000/\text{QALY}$ through the comparator (NT). All frontier segments are flatter than the WTP line, indicating cost-effectiveness at every threshold $\geq \approx \text{£}6,300/\text{QALY}$.

4.2 Net Monetary Benefit and the Recommendation

Translating the frontier into net monetary benefit ($\text{NMB} = \lambda \cdot \text{QALYs} - \text{Costs}$) collapses the multi-strategy comparison into a single ranking at each threshold (Stinnett & Mullahy, 1998), avoiding the error-prone exclusion logic required for fully incremental ICERs and giving a direct sense of the economic significance of each step.

Table 5. Net monetary benefit by strategy at six cost-effectiveness thresholds (£)

Threshold	NT	MM	EST	ANG	Optimal
£5,000	11,823	21,122	23,011	28,304	ANG
£10,000	26,025	49,564	51,476	51,753	ANG
£15,000	39,037	74,941	79,341	80,056	ANG
£20,000	52,049	100,318	107,207	108,360	ANG
£25,000	65,061	125,694	135,072	136,663	ANG
£30,000	78,074	151,071	162,938	164,967	ANG
£35,000	91,086	176,448	190,803	193,280	ANG

Angiography maximizes NMB at every threshold tested. The incremental net monetary benefit of ANG over MM i.e., $\text{INMB} (\text{ANG} - \text{MM})$ rises from $+\text{£}7,235$ at $\lambda = \text{£}5,000$ to $+\text{£}13,896$ at $\text{£}30,000$ and $+\text{£}16,822$ at $\text{£}35,000$. The economic gap between ANG and the second-best strategy, however, narrows sharply with the threshold: at $\lambda = \text{£}30,000$ the INMB of ANG over EST is only $+\text{£}2,029$, whereas the gap over MM is $+\text{£}13,896$. The practical reading is that once the decision-maker accepts ANG over MM, the marginal case for choosing ANG specifically

over EST is real but modest and depends on the value placed on the additional 0.088 QALYs that perfect diagnostic information delivers.

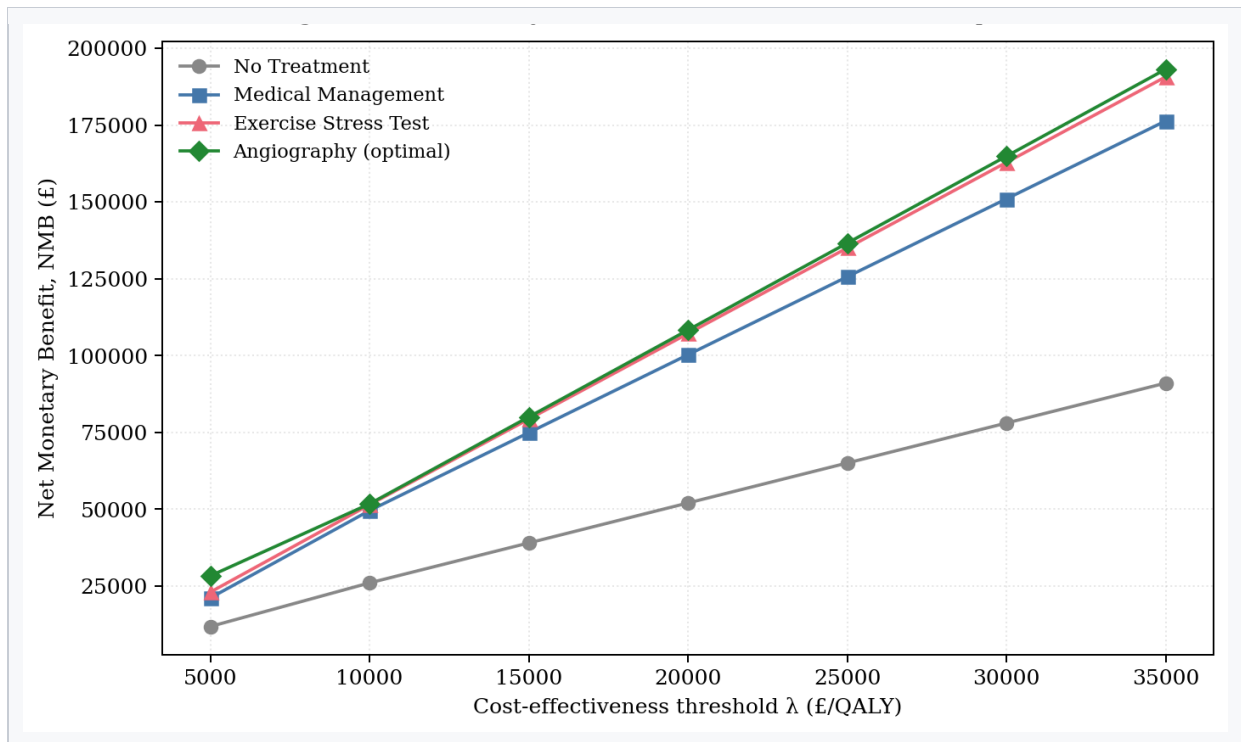


Figure 4. NMB across cost-effectiveness thresholds (base case). Lines for MM, EST, and ANG fan out roughly in parallel because they share most of the QALY mass; the vertical distance between ANG and MM widens with λ , reflecting the increasing monetary valuation of ANG's QALY gain. ANG remains the upper envelope across the full plausible UK range.

The base-case recommendation (RQ1) is therefore unambiguous: angiography is the cost-effective strategy at every threshold in the plausible UK range, with INMB over the next-best strategy ranging from +£330 (at £10,000) to +£2,520 (at £35,000), and INMB over the status quo of medical management ranging from +£7,235 to +£16,822. Whether the strength of that recommendation survives uncertainty in the key inputs is the question Section 5 addresses.

5 Sensitivity Analysis

A good decision model is judged not by the precision of its base-case point estimate but by whether its policy implication holds across the range of plausible parameter values (Kaltenthaler et al., 2011). This section interrogates the angiography recommendation along three axes: one-way deterministic sensitivity (parameter uncertainty), two-way sensitivity (joint uncertainty and population heterogeneity), and a value-of-improvement analysis that uses the NMB framework to quantify the return on investing in better surgical performance.

5.1 One-way Sensitivity: Which Parameters Move the Recommendation?

Figure 5 is a tornado diagram showing the change in INMB(ANG – MM) at $\lambda = £30,000$ when each parameter is varied across a plausible range with all others held at base case. Bars are ranked by total swing. Four parameters dominate: the cost-effectiveness threshold itself, CABG mortality in MVD patients, CABG success rate in MVD patients, and disease prevalence $p(S)$. Three results are worth flagging.

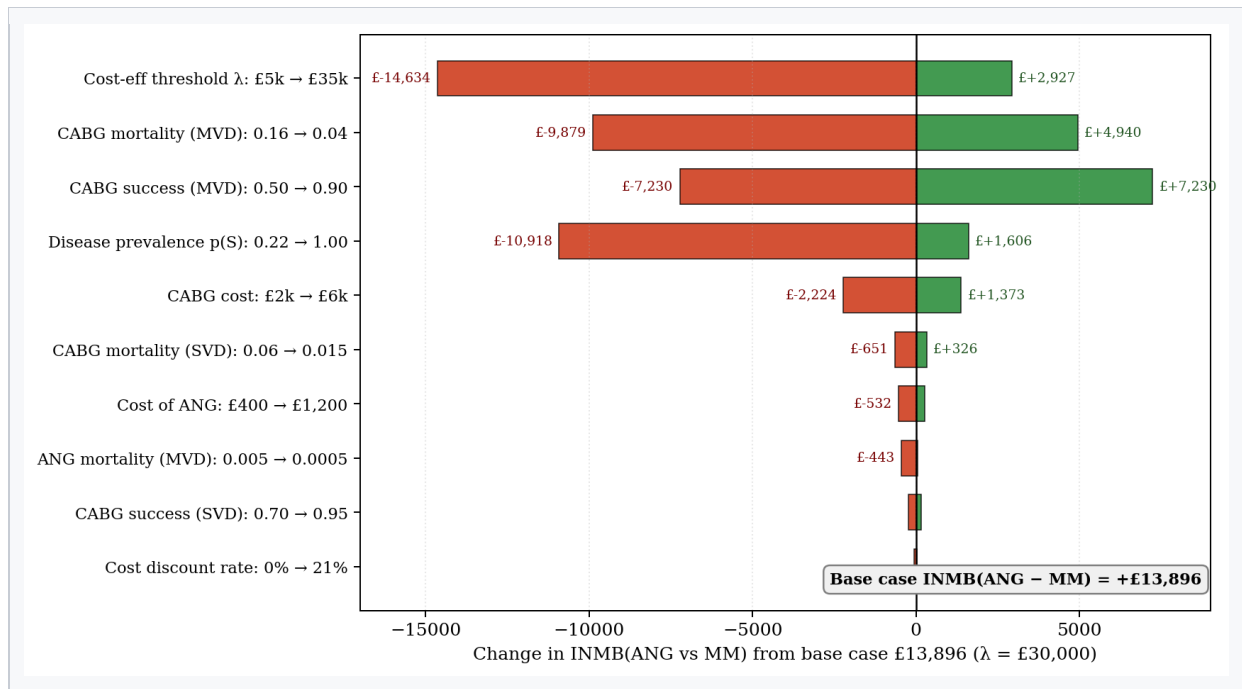


Figure 5. Tornado diagram - sensitivity of INMB(ANG - MM) at $\lambda = £30,000$.

First, the model is more sensitive to surgical performance than to diagnostic strategy. Raising MVD CABG mortality from 0.08 to 0.16 reduces INMB(ANG - MM) by $\approx £9,900$, and dropping MVD success from 0.70 to 0.50 reduces it by $\approx £7,200$. This is intuitive: ANG's advantage over MM is entirely the QALYs delivered by surgery, so the quality of surgery is what the recommendation truly rests on.

Second, EST sensitivity and specificity have zero effect on INMB(ANG - MM), because neither ANG nor MM uses the EST. The right comparator for EST parameters is the ANG-EST gap. At base case, INMB(ANG - EST) at $\lambda = £30,000$ is $£2,029$; raising EST sensitivity from 0.85 to 1.00 collapses this to $\approx £0$, meaning a perfect EST would essentially tie with ANG; dropping it to 0.70 widens the gap to $£4,123$. EST specificity barely matters ($\pm £20$). This is a cleaner reading than treating EST parameters as drivers of the headline result.

Third, the QALY discount rate has limited influence on the recommendation. ANG remains optimal at $\lambda = £30,000$ for QALY discount rates up to approximately 12–13%, well above the 3.5% NICE reference rate (NICE, 2022), because most of ANG's QALY gain accrues in the first 5–10 years. The cost discount rate moves INMB by less than $£50$ across the full 0%–21% range.

Figure 6 plots NMB by strategy as p(S) varies at $\lambda = £10,000$. ANG overtakes MM at a break-even prevalence of p(S) ≈ 0.22 , well below the Sheffield value of 0.90 and below the prior probability for moderate angina (0.60). Equivalently, holding p(S) at base case, ANG remains optimal at every $\lambda \geq £6,232/\text{QALY}$.

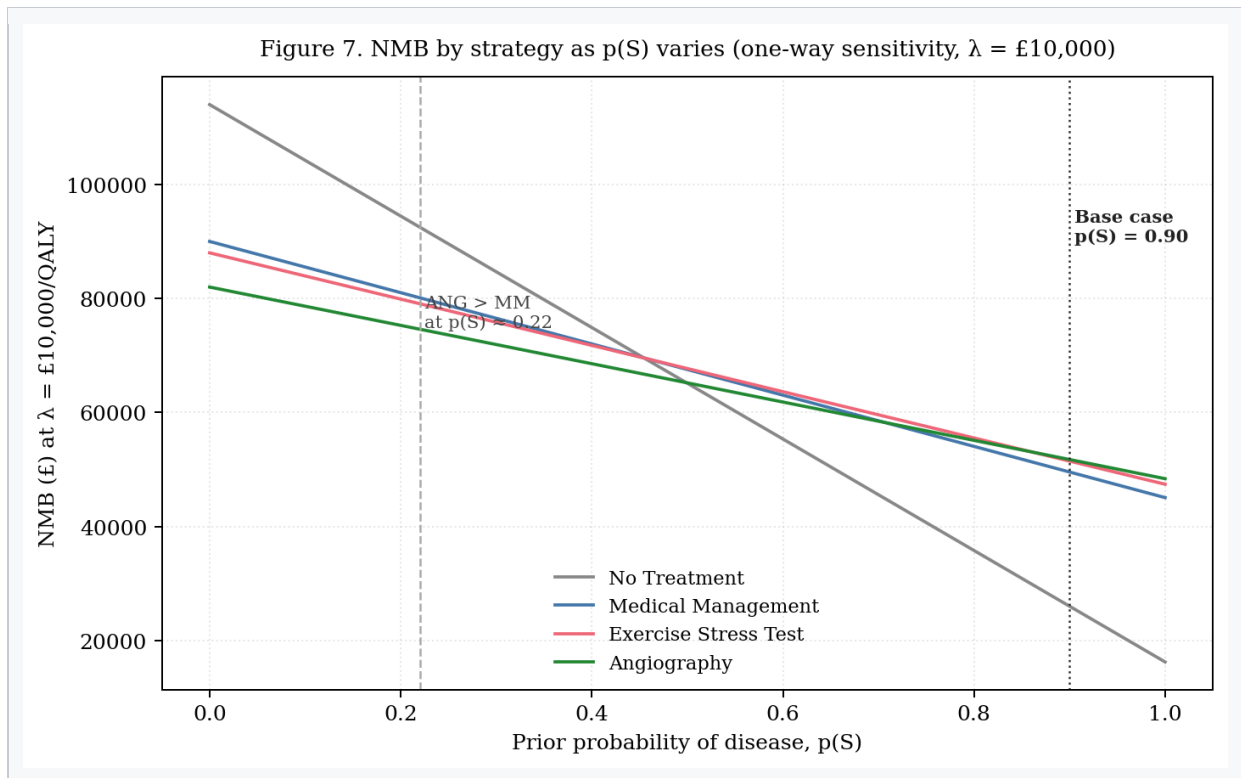


Figure 6. NMB by strategy as $p(S)$ varies at $\lambda = £10,000$. ANG overtakes MM at $p(S) \approx 0.22$; the Sheffield base case ($p[S] = 0.90$) sits far inside the ANG-optimal range.

5.2 Two-way Sensitivity: Heterogeneity and the Decision Surface

Figure 7 maps the optimal strategy across the full prevalence $\times$ threshold grid. Four regions emerge: NT is optimal only at $p(S) = 0$; MM dominates a low-prevalence, low-threshold band; EST occupies a narrow transitional region; and ANG fills the upper-right quadrant of high prevalence and/or high threshold.

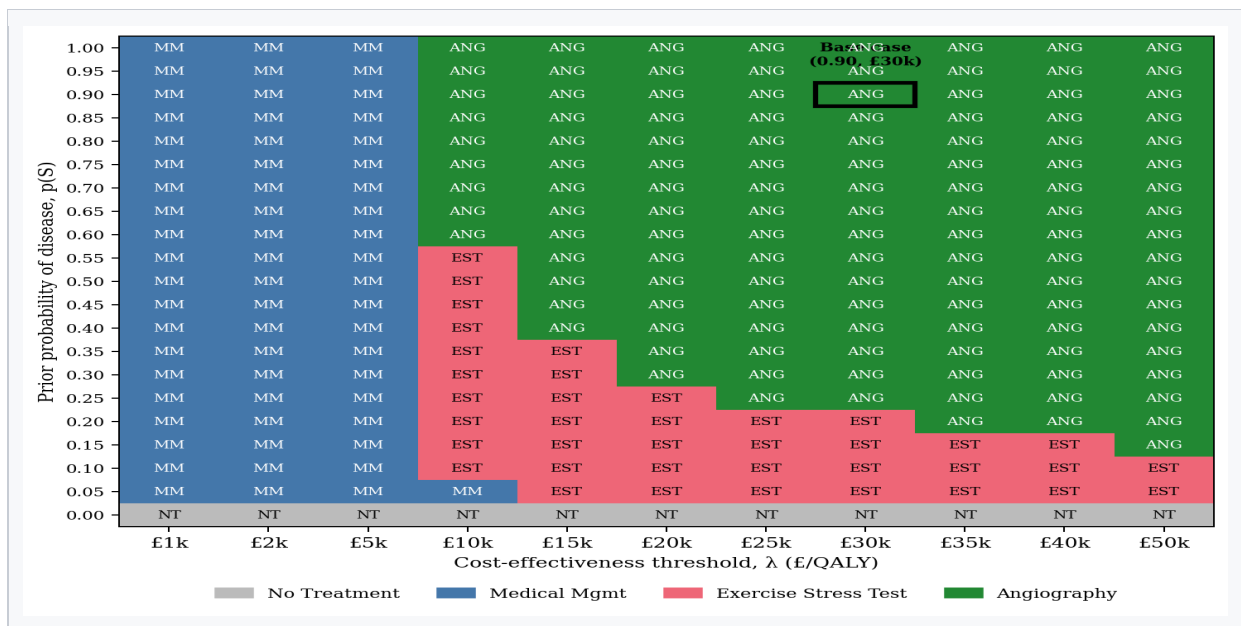


Figure 7. Optimal strategy heatmap across the prevalence ($p[S]$) $\times$ cost-effectiveness threshold (λ) grid. The Sheffield base case ($p[S] = 0.90$, $\lambda = £10,000$) sits deep within the ANG-optimal region.

The Sheffield base case sits deep within the ANG region, three rows and four columns inside the boundary. For the recommendation to flip to MM, both prevalence and threshold would need to fall simultaneously to roughly $p(S) \leq 0.05$ and $\lambda \leq £5,000$.

Heterogeneity. This grid is also the heterogeneity analysis. The same model can be re-read as comparing distinct patient populations: severe-angina referrals ($p[S] = 0.90$) sit in the deep-ANG region; moderate-angina referrals ($p[S] = 0.60$) sit just inside the ANG region at NICE thresholds but on the EST/ANG boundary at lower thresholds; asymptomatic screening populations ($p[S] < 0.10$) sit firmly in the MM region. The same decision tree therefore yields different recommendations for different patient mixes — a feature, not a bug, of the modelling framework.

EST diagnostic performance. A separate grid over EST sensitivity and specificity confirms ANG is optimal across virtually the entire parameter space; EST becomes optimal only at near-perfect sensitivity and specificity ≈ 0.80 .

5.3 Value of Improvement: Return on Better Surgery

Decision models earn their keep not only by ranking existing strategies but by valuing potential improvements to them. The NMB framework translates parameter improvements directly into pounds, the maximum a regulator should be willing to invest. Figure 8 plots INMB(ANG – MM) at $\lambda = £30,000$ as the CABG success rate for MVD patients varies from 0.40 to 1.00.

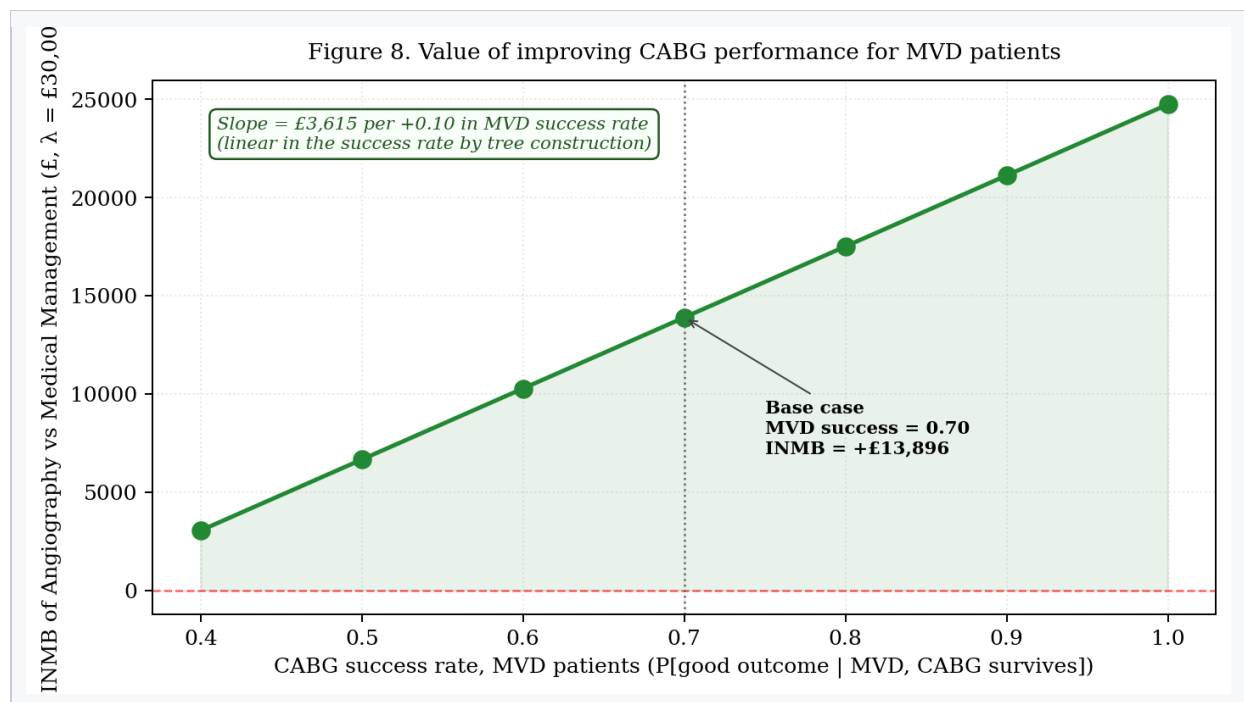


Figure 8. Value of improving CABG performance. The slope is £3,615 in INMB per +0.10 in MVD success rate, linear by the tree's structure.

For a cardiology department running approximately 200 severe-angina CABG cases annually, an intervention raising MVD success from 0.70 to 0.80 (one absolute decile) would generate roughly £723,000 in net monetary benefit per year, a defensible upper bound on what the department should be willing to spend on such an intervention. The general principle is that returns to investment are largest where the parameter sits in the active part of the decision surface improving CABG (the treatment ANG delivers) pays back far more than improving EST (a comparator already off the frontier).

5.4 Transferability Scenario: Lower-threshold Settings

Below approximately £6,232/QALY, a threshold range relevant to lower-middle-income systems and consistent with WHO-CHOICE opportunity-cost estimates for several LMICs, the heatmap (Figure 7) shows MM becoming optimal across the prevalence range up to $p(S) \approx 0.50$, with EST occupying the transitional band as a capacity-conserving triage tool where catheterisation is scarce. The structural conclusion that angiography is justified at high prevalence survives, but the prevalence required to justify it rises sharply. A full transferability assessment would re-estimate local unit costs and CABG mortality.

5.5 Limits of the Deterministic Approach

Probabilistic sensitivity analysis (PSA) was not undertaken because no distributional information was provided for the input parameters. Were PSA conducted with plausible distributions, the cost-effectiveness acceptability curve at $\lambda = £30,000$ would almost certainly show ANG optimal in the substantial majority of simulations, given that the deterministic INMB(ANG – MM) is +£13,896 and survives all single-parameter perturbations except extreme $p(S)$ reductions.

6 Discussion

6.1 Answering the Research Questions

The base-case analysis (RQ1) identifies angiography as the cost-effective strategy for the Sheffield severe-angina cohort at every threshold in the plausible UK range, including values below the NICE reference range that more credibly reflect NHS opportunity cost (Claxton et al., 2015). The INMB of ANG over MM is +£13,896 at $\lambda = £30,000$ and remains positive down to $\lambda \approx £6,232/\text{QALY}$, well below any defensible UK threshold. The recommendation is therefore not threshold-contingent within the relevant policy range.

The two-way analysis (RQ2) locates the Sheffield cohort deep inside the ANG-optimal region of the prevalence $\times$ threshold surface. For the recommendation to flip to MM, both $p(S)$ and λ would need to fall simultaneously to approximately 0.05 and £5,000, a combination outside any plausible Sheffield scenario.

On the value of diagnostic accuracy (RQ3), the analysis yields a counter-intuitive but defensible result: EST sensitivity and specificity barely move the headline recommendation, because EST is already off the frontier in the comparison that matters. The marginal return to improving EST is approximately £1,380 per +0.10 sensitivity per patient, against £3,615 per +0.10 in CABG success; surgery, not diagnosis, is where investment pays back. On structural robustness and transferability (RQ4), the recommendation survives QALY discount rates up to 12–13% and a relaxation of the perfect-angiography assumption, but does not survive a simultaneous threshold and prevalence collapse, the scenario relevant to lower-income settings.

6.2 Implications of Assumptions and Limitations

Four structural choices deserve explicit treatment for the decision-maker. Static disease states (assumption i) biases the model against ANG by suppressing the long-term QALY gains from earlier accurate diagnosis; the recommendation is therefore conservative on this dimension, and a Markov extension would, if anything, strengthen it. Perfect angiography (assumption ii) biases in the opposite direction, but the +£13,896 INMB cushion over MM is large enough that empirically plausible ANG accuracy ($\approx 0.95\text{--}0.99$) leaves the conclusion intact. Rosser–Kind utilities (assumption iii) likely inflate absolute QALY totals modestly but affect strategy differences less, so the ranking is robust. The absence of PSA (assumption iv) is the most

consequential limitation: although the deterministic surface gives reasonable confidence that ANG would dominate the cost-effectiveness acceptability curve, the residual decision uncertainty has not been quantified and should be the priority for any follow-up.

6.3 Policy Implication

For the Sheffield department, the analysis supports routine direct angiography for severe-angina referrals at the current cohort prior. Where catheterisation capacity is constrained, EST retains value as a triage tool, but the analysis suggests the larger return on departmental investment lies in improving CABG outcomes in MVD patients rather than in upgrading the diagnostic pathway.

7 Conclusion

Direct angiography is the cost-effective strategy for the Sheffield severe-angina cohort across the full plausible UK threshold range, with the recommendation robust to all single-parameter perturbations except a simultaneous collapse in prevalence and threshold. Priorities for further work are probabilistic sensitivity analysis with elicited input distributions, and a Markov extension to capture progression dynamics.

8 References

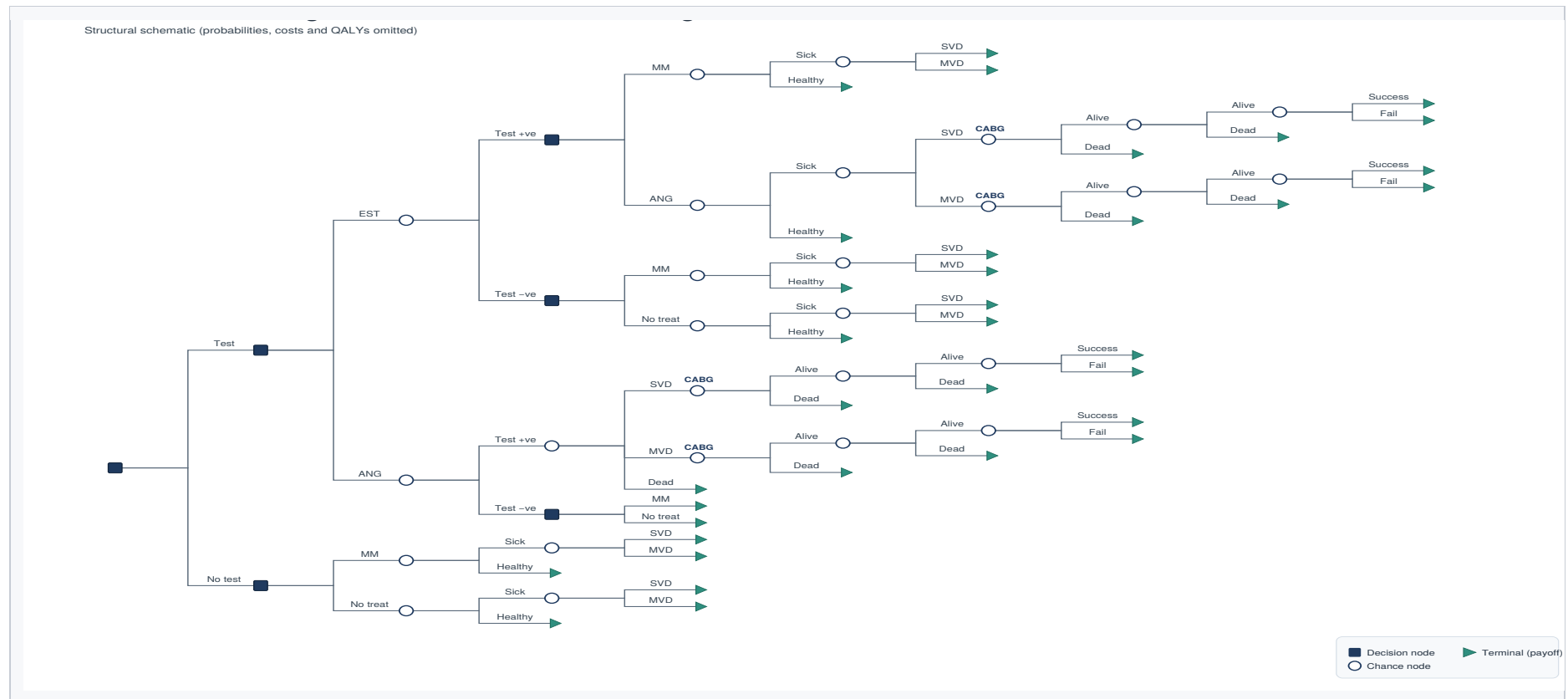
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Appendices

The appendices hold supporting material that underpins the analysis but is not essential to the main argument. Space is reserved below for figures and tables to be inserted from the model workbook prior to submission.

Appendix A Full Decision-Tree Structure

Complete, unfolded decision tree for all four strategies, with chance-node probabilities, terminal-nodes.



Appendix B State-by-State QALY Derivation

Year-by-year Rosser–Kind health-state assignments and discounted QALY accumulation for each disease state and treatment pathway (severe-pain cohort), supporting the summary figures in Table 3.

Pathway	State	Utility Weights	Years in State	Discounted QALY contribution	Pathway Total
SVD → CABG (success) → MM	IA	1.000	Year 1	0.966	
	IIC	0.973	Year 2-3	1.786	
	IIIC	0.956	Years 4-10	5.408	8.16
MVD → CABG (success) → MM	IB	0.995	Year 1	0.961	
	IIA	0.990	Year 2-3	1.817	
	IVC	0.942	Years 4-10	2.822	5.60
SVD → MM (no surgery)	IIB	0.986	Year 1-3	2.760	
	IVC	0.942	Year 4-7	3.024	
MVD → MM (no surgery)	VD	0.700	Years 8-10	1.826	7.61
	IIIC	0.956	Year 1-2	1.817	
	IVD	0.870	Year 3-6	2.193	
SVD → No treatment	VD	0.700	Years 7-10	0.000	4.01
	VD	0.700	Year 1-5	3.161	
	VID	0.000	Year 6-10	1.119	4.28

Pathway	State	Utility Weights	Years in State	Discounted QALY contribution	Pathway Total
MVD → No treatment	VD	0.700	Year 1-2	1.331	1.33
	VID	0.000	Year 3-5	0.000	
	(Death)	0.000	Years 6-10	0.000	
Healthy — any strategy	IA	1.000	Years 1-5	4.515	11.40
	IIB	0.986	Years 6-15	6.885	

Appendix C: Consolidated Parameter Table with Sensitivity Ranges and Sources

Table C1 consolidates all model inputs used in the base-case analysis (Section 4) and the one-way deterministic sensitivity analysis (Section 5.1). Bounds for the latter are drawn from clinically plausible ranges in the Sheffield case study unless otherwise indicated.

Table C1. Full model parameter set. All monetary values in GBP. Bounds defined as the ranges used in one-way deterministic sensitivity analysis.

Parameter	Base Case	Low	High	Source/basis
Probability Inputs				
Prior probability of disease, p(S)	0.90	0.00	1.00	Sheffield case study; full range explored for heterogeneity
P(SVD sick)	0.10	0.05	0.20	Sheffield case study
P(MVD sick)	0.90	0.80	0.95	Sheffield case study
EST sensitivity, P(+ sick)	0.85	0.70	1.00	Sheffield case study; range reflects published EST performance
EST specificity, P(– healthy)	0.90	0.80	1.00	Sheffield case study
ANG sensitivity / specificity	1.00 / 1.00	0.95 / 0.95	1.00 / 1.00	Structural assumption; relaxed in Section 5.3
CABG success rate — SVD	0.85	0.75	0.95	Sheffield case study
CABG success rate — MVD	0.70	0.50	0.90	Sheffield case study; range per Section 5.1
Procedure mortality — CABG (SVD)	0.03	0.01	0.05	Sheffield case study

Parameter	Base Case	Low	High	Source/basis
Procedure mortality — CABG (MVD)	0.08	0.04	0.16	Sheffield case study; range per Section 5.1
Procedure mortality — ANG	0.0003	0.0001	0.001	Sheffield Case Study
Procedure mortality — no disease	0.0002	0.0001	0.0005	Sheffield case study
Cost inputs (GBP)				
Exercise stress test (EST), unit cost	53	40	80	Sheffield case study
Angiography (ANG), unit cost	668	500	900	Sheffield case study
CABG procedure, unit cost	3,527	2,800	4,500	Sheffield case study
MM — post-CABG success, annual	107	80	150	Sheffield case study
MM — severe / CABG failure, annual	214	160	300	Sheffield case study
MM — healthy / no treatment, annual	214	160	300	Sheffield case study
No treatment	0	-	-	By definition
Utility inputs (QALY weights)				
Rosser–Kind utility weights	See Appendix B	-	-	Kind, Rosser & Williams (1982); NICE reference standard is EQ-5D (see Section 3.5 and 6.2)
Discount rates				

Parameter	Base Case	Low	High	Source/basis
Cost discount rate	3.5%	0%	21%	NICE reference case (NICE, 2022)
QALY discount rate	3.5%	0%	21%	NICE reference case (NICE, 2022)
Decision rule				
Cost-effectiveness threshold, λ	£20,000–£30,000	£5,000	£35,000	NICE reference range (NICE, 2022); lower bounds reflect Claxton et al. (2015) opportunity-cost estimate
Time horizon				
Post-CABG success	10 years	-	-	Sheffield case study
Conservative / healthy / no treatment	15 years	-	-	Sheffield case study

Appendix D: Bayesian Posterior Calculations for the Exercise Stress Test

The EST arm of the decision tree updates the prior probability of disease, $p(S)$, based on the test result, using Bayes' theorem. The posterior probabilities derived here govern the conditional treatment decision following EST in the model.

Inputs (base case)

Prior probability of disease: $p(S) = 0.90$

Prior probability of no disease: $p(\neg S) = 1 - p(S) = 0.10$

EST sensitivity: $P(\text{EST}+ \mid S) = 0.85$

EST specificity: $P(\text{EST}- \mid \neg S) = 0.90$

False positive rate: $P(\text{EST}+ \mid \neg S) = 1 - 0.90 = 0.10$

False negative rate: $P(\text{EST}- \mid S) = 1 - 0.85 = 0.15$

Step 1: Marginal probability of a positive test result

By the law of total probability:

$$P(\text{EST}+) = P(\text{EST}+ \mid S) \times p(S) + P(\text{EST}+ \mid \neg S) \times p(\neg S)$$

$$P(\text{EST}+) = (0.85 \times 0.90) + (0.10 \times 0.10) = 0.765 + 0.010 = 0.775$$

Step 2: Marginal probability of a negative test result

$$P(\text{EST}-) = P(\text{EST}- \mid S) \times p(S) + P(\text{EST}- \mid \neg S) \times p(\neg S)$$

$$P(\text{EST}-) = (0.15 \times 0.90) + (0.90 \times 0.10) = 0.135 + 0.090 = 0.225$$

Step 3: Posterior probability of disease given a positive test (positive predictive value)

By Bayes' theorem:

$$P(S \mid \text{EST}+) = [P(\text{EST}+ \mid S) \times p(S)] / P(\text{EST}+)$$

$$P(S \mid \text{EST}+) = 0.765 / 0.775 = 0.9871$$

Step 4: Posterior probability of disease given a negative test

$$P(S \mid \text{EST}-) = [P(\text{EST}- \mid S) \times p(S)] / P(\text{EST}-)$$

$$P(S \mid \text{EST}-) = 0.135 / 0.225 = 0.6000$$

Table D1. Summary of EST posterior probabilities at base case.

Test result	Posterior $P(S \mid \text{result})$	Posterior $P(\neg S \mid \text{result})$	Marginal $P(\text{result})$
EST positive (+)	0.9871	0.0129	0.7750
EST negative (-)	0.6000	0.4000	0.2250